

SHARPENING CHEAT SHEET

Angles, grits and strops — by knife type. Numbers are per side, measured off the stone. Laminate this page and hang it where you sharpen.

EDGE ANGLE BY KNIFE TYPE

KNIFE	PER SIDE	WHY
Straight razor	7–9°	Pure slicing, zero lateral load
Fillet / sushi	12–15°	Effortless slicing in soft flesh
Kitchen chef / paring	15–20°	Sharpness first; the steel sets how low you dare
EDC / pocket	17–20°	Daily cutting + some abuse
Hunting / skinner	16–20°	Start at 18°. Premium steel holds 16° through bone
Bushcraft / camp	18–25°	Batoning, carving — harder steels run the low end
Machete / axe	25–30°	Impact first, sharpness second

The steel sets the floor: better steel = lower angle for the same job. A CruWear, 3V or MagnaCut hunter holds 16° and hits bone with no problem; budget stainless wants 20°. When in doubt: start 20°, work down as the steel earns your trust.

HOLD THE ANGLE — THREE TRICKS

Coin stack	Two quarters under the spine of a 1.5" blade ≈ 15°. Cheap training wheels.
Sharpie trick	Ink the bevel, two passes, look where ink is gone. Adjust until the full bevel wipes clean.
Lock the wrist	Move from the shoulder. The wrist changes angle; the shoulder doesn't.

GRIT LADDER

STAGE	GRIT	USE WHEN
Repair	220–400	Chips, rolled edge, reprofiling the bevel
Sharpen	600–1000	The workhorse stage. Raise a burr full length, both sides
Refine	2000–3000	Clean up scratches, chase the burr smaller
Polish	5000–8000	Push-cut hair-whittling territory. Kitchen + razors
Strop	compound	Every session ends here. Also revives an edge for weeks

Skip grits, waste time: each stage must fully erase the scratches of the one before. Doubling the grit number per step is the fastest ladder.

STROP COMPOUNDS

COMPOUND	~MICRON	JOB
Black (emery)	~30μ	Heavy cutting, almost a stone
White (alum. oxide)	~3μ	Mid polish, daily maintenance
Green (chrom. oxide)	~0.5μ	Finish polish, scary sharp
Bare leather	—	Aligns the edge, removes nothing

THE 5-STEP ROUTINE

1	Mark the bevel with a Sharpie. Set your angle.
2	Low grit until a burr runs the FULL edge on one side. Flip, repeat.
3	Climb the ladder. Lighter pressure every stage.
4	Strop 10–15 passes per side, edge trailing, angle slightly low.
5	Test: slice printer paper, then shave arm hair. Fail → back one grit.

STONE TYPES, HONESTLY COMPARED

STONE	CUTS	TRADE-OFF
Water stones	Fast, fine feedback	Dish quickly — flatten often. The learning stone.
Diamond plates	Everything, incl. PM steels	Never dish, cut forever — scratch deeper than the grit says.
Ceramic rods	Touch-ups, serrations	Maintenance, not repair. Clean with eraser.
Oil stones	Carbon steels, slow + sure	Slow on modern stainless; struggle on PM carbides.
Belt + leather wheel	Fastest by a mile	Heat kills tempers — light passes, dunk often, edge NEVER burns blue.

PM steels (MagnaCut, 3V, CruWear) want diamonds. Their carbides are harder than aluminum-oxide stones — soft stones polish AROUND the carbides and the edge never really finishes.

THE BURR — THE WHOLE GAME IN THREE LINES

Raise it	Grind one side until a wire of steel folds over the FULL length. Feel it with a fingernail dragged off the edge.
Chase it	Flip, repeat, burr flips sides. Each grit: lighter pressure, smaller burr.
Kill it	Final edge-leading feather passes, then strop. A left-over burr feels sharp for one cut, then folds and "goes dull."

ANGLE FLOORS BY STEEL CLASS

STEEL CLASS	FLOOR	EXAMPLES
Budget stainless	20°	420HC, 8Cr13MoV, mystery stainless
Standard carbon	18°	1075 · 1084 · 1095 · 80CrV2
Fine-grain carbon	17°	52100 · 26C3 · W2
Quality stainless	16–17°	AEB-L · 14C28N · Nitro-V
Premium PM	16°	CruWear · CPM-3V · MagnaCut — bone, no problem

Floor = lowest sensible angle with proper heat treatment. Above the floor is always safe; below it the edge folds or chips in real use.

TROUBLESHOOTING

SYMPTOM	FIX
Won't get sharp at all	You never raised a full-length burr. Go back to the low grit and stay there.
Sharp for one day	Burr never killed — it folded. Feather passes + strop, lighter every stroke.
Edge chips in use	Angle below the steel's floor, or the blade runs too hard. Add 2–3°.
Edge folds / rolls	Soft steel or too-low angle: raise angle; steel the roll back first.
Shiny line on the very edge	You're not reaching the apex — angle too low on the stone. Sharpie trick, adjust up.
Scratches won't polish out	You skipped a grit. Ladder discipline: double the number each step.

KEEP IT SHARP — THE SCHEDULE

Every use	Wipe dry. Carbon steel: thin coat of oil.
Weekly (kitchen)	10 strop passes per side. 30 seconds, saves an hour.
When strop stops helping	Back to the 1000 grit — a real apex reset.